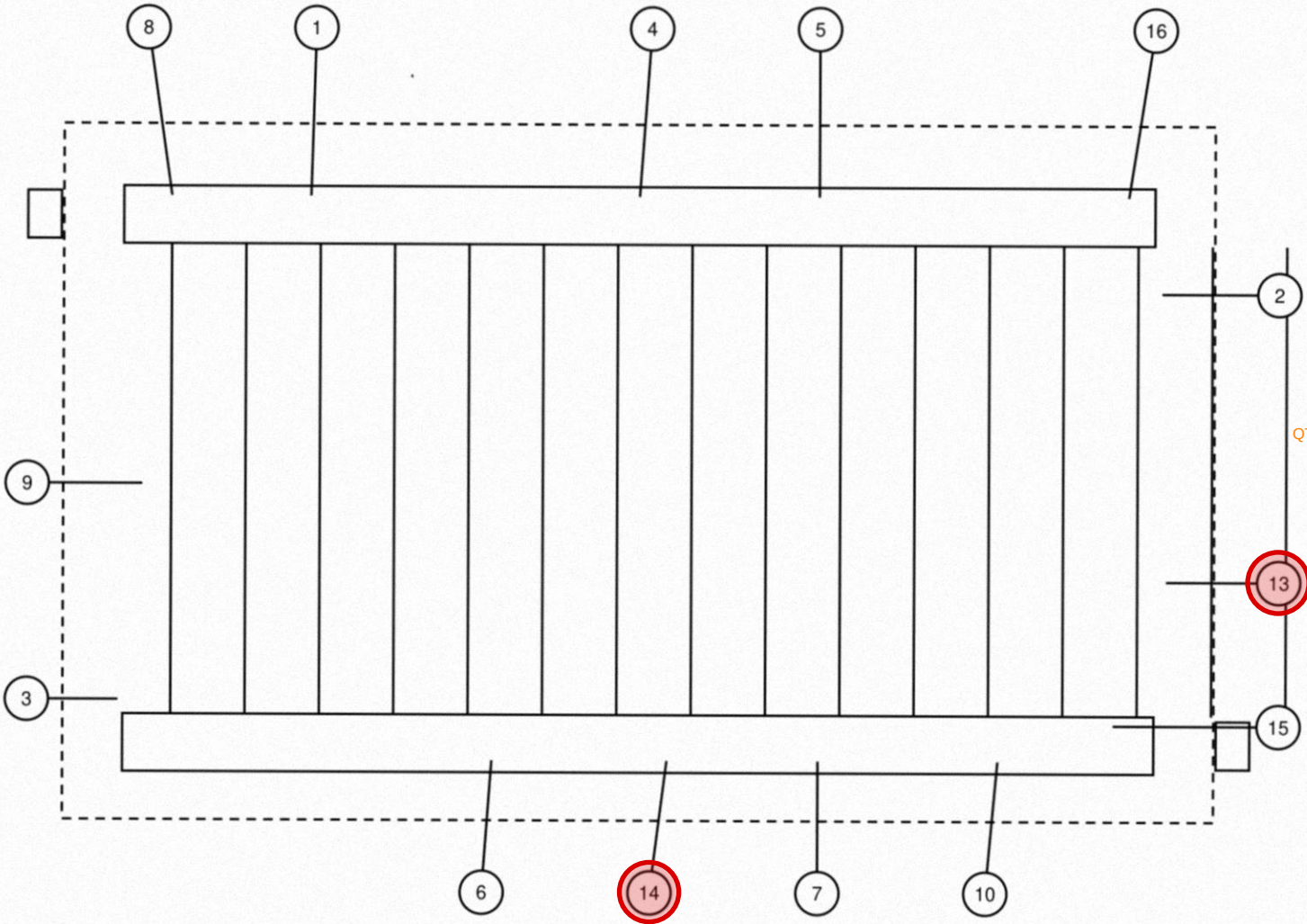


BOM CHECK (automated - verify before acting)

- red = on drawing, missing in BOM
- orange = qty / spec / revision mismatch
- yellow = confirm match / extra / duplicate
- blue = suggestion (implied item, rule-based)



SPEC MISMATCH

QTY: BOM 116, expected 464

QTY: dwg 2 / BOM 3

MISSING IN BOM

MISSING IN BOM

Sr No	Part Description	Material	Size	Qty
1	INLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B		1
2	OUTLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B		1
3	ECONOMISER TUBE 38.1 OD X 3.6 THK	SA 210 GR.A1		116
4	TUBE SUPPORT PLATE 12 THK	IS 2062 GR.B		14
5	INLET FLANGE 200 NB CL.150	SA 105		1
6	OUTLET FLANGE 200 NB CL.150	SA 105		1
7	SAFETY VALVE 50 NB	SA 216 WCB		1
8	DRAIN VALVE 25 NB	SA 105		2
9	VENT VALVE 15 NB	SA 105		1
10	THERMOWELL 1/2 NPT	SS 316		2
11	GASKET 200 NB CL.150	SS 304 + GRAPHITE		2
12	STUD BOLT M20 X 90 WITH 2 NUTS	SA 193 B7		16
13	LIFTING LUG 16 THK	IS 2062 GR.B		4
14	CASING PLATE 3 THK	IS 2062		23
15	NAME PLATE	SS 304		1
16	INSPECTION DOOR 300 X 300	IS 2062		2

NOTES:

- ALL DIMENSIONS ARE IN MM.
- ALL **WELDING** AS PER IBR / ASME SEC I. <- BOM LACKS: Welding consumables
- INSULATE WITH 100 THK MINERAL WOOL.
- PAINT: ONE COAT RED OXIDE PRIMER.
- TUBE QTY IS PER BANK. TYPICAL FOR 4 BANKS.

ECONOMISER ASSEMBLY

DRG NO : EC-03-GA-001
DRG REV : D
SCALE : NTS SIZE : A3
SAMPLE (SYNTHETIC) DRAWING

BOM CHECK SUMMARY

Drawing : sample_03_drawing_scan.pdf (Rev D)
BOM : sample_03_bom.xlsx (Rev D)
Input : scanned (OCR)
RUN RELIABILITY : HIGH

Drawing items checked : 16
BOM lines read : 17
BOM completeness : 87.5% (drawing items that have a BOM line)

Findings: extra 1, implied 1, multiplier 1, omission 2, quantity 1, spec 1

[RED] Item 13 'LIFTING LUG 16 THK' (qty 4) is on the drawing but has no BOM line
[RED] Item 14 'CASING PLATE 3 THK' (qty 23) is on the drawing but has no BOM line
[ORANGE] Item 1: specification mismatch - dimension: drawing ['8'] vs BOM ['10']
[ORANGE] Item 3: drawing note says qty is per bank, x4; BOM has 116 but total should be 464
[ORANGE] Item 11: quantity differs (drawing 2, BOM 3)
[YELLOW] BOM row 20 'Hand rail, 42 NB' has no matching item on the drawing (extra, or from another revision)
[BLUE] Suggestion (Welding consumables): the drawing notes imply it, but no matching line was found in the BOM

Automated check - not a substitute for engineering review. Blue items are rule-based suggestions.